

1 Superposition Codes

We begin by describing the toy scenario to be studied. Given a large number N , consider a map F that “encodes” each subset $y \subseteq [N] = \{1, \dots, N\}$ by a linear combination

$$x = \sum_{i \in y} f_i \in \mathbb{R}^d,$$

where the vectors $\{f_i \in \mathbb{R}^d : i \in [N]\}$ are chosen in advance and where the dimension d of the encoding is expected to be much smaller than N . We call the vectors f_i codewords for the elements of $[N]$ and call the image Fy a superposition code for the set y . It is useful to view y as a vector in $\{0, 1\}^N$ with coefficients

$$y_i = \begin{cases} 1 & : i \in y \\ 0 & : \text{otherwise} \end{cases}$$

and view F as a matrix of column vectors $[f_1 \dots f_N]$, called the dictionary. In this work, we’ll model our subset as a random variable Y uniformly distributed over all subsets of some fixed size $k \ll N$.

One natural approach to designing the dictionary F is to choose the codewords f_i randomly. This is motivated by the fact that, so long as $d = \Omega(\ln N)$, we can expect random codewords to be nearly orthogonal. More precisely, we have results like the following.

Proposition 1 *For $\epsilon > 0$ and $p > 0$, let $d > 2\epsilon^{-2}(2 \ln N + \ln p^{-1})$ and let*

$$\{F_1, \dots, F_N\} \subseteq \{-1/\sqrt{d}, 1/\sqrt{d}\}^d$$

be random vectors with independent, uniformly distributed entries. Then

$$\forall i \neq j, |\langle F_i, F_j \rangle| < \epsilon \tag{1}$$

with probability at least $(1 - p)$.

If we put $\epsilon = 1/2k$, then the condition (1) guarantees that y can be decoded from the superposition code Fy in one obvious way: specifically, by applying the thresholding rule

$$\sigma(t) = \begin{cases} 1 & : t \geq 1/2 \\ 0 & : \text{otherwise} \end{cases}$$

element-wise to the vector $F^T Fy$. Proposition 1 thus guarantees that, to encode sets of a fixed size k , superposition codes require only $O(\ln N)$ dimensions asymptotically in N .

One-step methods can be derived by separately considering the estimation problem for each coordinate Y_i

$$X = Y_i f_i + \overbrace{\sum_{j \neq i} Y_j f_j}^Z \tag{2}$$

and approximating the sum Z of other codewords as Gaussian noise. Under this simplification, it is easy to infer Y_i ; if we model Z as a centered Gaussian with non-singular covariance matrix Σ and define an inner product by $\langle v, w \rangle_\Sigma = 1/2 v^T \Sigma^{-1} w$, then the exact posterior of Y_i conditional on X is determined by the function

$$\lambda_i(X) = \frac{\langle f_i, X \rangle_\Sigma}{\|f_i\|_\Sigma^2}.$$

In signal processing, this kind of linear measurement is called a **matched filter** for the codeword f_i . The log odds of the posterior on Y_i in terms of the matched filter is given by

$$\begin{aligned} & \ln \frac{P(Y_i = 1 | X = x)}{P(Y_i = 0 | X = x)} \\ &= \rho \left(\lambda(x) - \frac{1}{2} \right) + \ln \frac{P(Y_i = 1)}{P(Y_i = 0)}, \end{aligned} \tag{3}$$

where $\rho = \|f\|_{\Sigma}^2$ is called the “signal-to-noise ratio” of the filter λ . See ?? for a review.

When the number k of non-zero coefficients is known, it is also natural to set $\hat{Y}_i = 1$ for the k indices i with maximum matched filters. This is called top- k decoding. Gao et al. (2024) showed that, in practice, top- k autoencoders perform better than their ReLU variants in practice.

The second class are the iterative methods.

We’re interested in understanding how large the dimension d needs to be as a function of (N, k) for a given inference method to recover Y . (We do not study the problem of learning the dictionary.) Since Y is a discrete variable, we will focus on conditions for *exact* recovery. We focus on a regime where Y resembles the very sparse latent representations learned by sparse autoencoders trained on large language models. For example, typical values discussed in Gao et al. (2024) are $N = 2^6$ and $k = 100$.

2 Information Theory Bounds

We recall the idea of channel capacity with a simple example. Suppose that the latent variable Y to be encoded is a symbol drawn from an alphabet of size N . What is the minimum dimension d such that each value of Y can be represented by a vector $X \in \mathbb{R}^d$?

When the vector X can be stored exactly, the answer to our question simply depends on the number of values it can take. If the coefficients of X are 16 bit floating point numbers, then each coefficient can store (nearly) 2^{16} distinct numbers, and overall we need only about $d_{\min} \approx \frac{1}{16} \log_2 N$ dimensions.

However, we expect that activation vectors within real neural networks are subject to various kinds of interference. For example, an activation vector X may need to represent the symbol Y while also holding information on another latent quantity, or it may be subject to explicitly induced noise like dropout. Characterizing the “capacity” of a vector in the presence of interference is a basic problem in coding theory.

One common model is to consider that X is subject to additive white Gaussian noise. More specifically, we let $Z \in \mathbb{R}^d$ be a vector of independent, centered Gaussians each of variance P , and consider the problem of recovering Y from $X + Z$. The following is then a typical (but remarkable) result of information theory.

Proposition 2 *Let the random variable $Y \in [N]$ be uniformly distributed, and let Z be white Gaussian noise of variance P . Suppose there exist a pair of maps $F: [N] \rightarrow \mathbb{R}^d$ and $G: \mathbb{R}^d \rightarrow [N]$ so that*

$$G(F(Y) + Z) = Y$$

with probability at least $(1 - p)$, and suppose the variance of each coordinate of $F(Y)$ is bounded by 1. Then

$$d \geq 2 \frac{(1 - p) \ln N - \ln 2}{\ln(1 + P^{-1})}.$$

In particular, if we fix P and let the number N of symbols grow to infinity, this shows that the minimum dimension d of a reliable code must grow logarithmically in N . More specifically, we can say the following.

Corollary 1 *Fix $P > 0$, $\epsilon > 0$, and suppose*

$$d \leq \left(\frac{2}{\ln(1 + P^{-1})} - \epsilon \right) \ln N.$$

Then, when the random variables (Y, Z) and the maps (F, G) are in the conditions of Proposition 2, the maximum value of $\mathbb{P}(G(F(Y) + Z) = Y)$ converges to 0 for sufficiently large N .

Since $\ln(1 + P^{-1})^{-1} \approx P$ for large P , this means that roughly $d \geq 2P \ln N$ dimensions are needed to reliably code for Y .

Now, one very simple way to construct maps F and G is to choose the codewords $F(i)$ randomly from some distribution on the unit sphere in $\mathbb{R}^d$ and let $G(v)$ return the element j so that $\langle v, F(j) \rangle$ is maximized. For

brevity, we refer to (F, G) . This strategy can be motivated by the remarkable fact that random codebooks of size $\Omega(e^d)$ of d -dimensional codewords can have bounded interference, in the sense of the following result.

Proposition 3 *Let (F, G) be constructed according to the random codebook strategy, let $\epsilon > 0$, and let*

$$d \geq (2 + \epsilon)P \ln N.$$

Then, when the random variable (Y, Z) are in the conditions of Proposition 2, $P(G(F(Y) + Z) = Y)$ converges to 1 for large N .

Altogether, this shows that transmitting

Our discussion so far parallels many Cover & Thomas (2006). In the remainder of this work, we consider

It will also be useful to know the upper bound

$$H = \ln \binom{N}{k} \leq k \ln(eN/k) = k \ln N - k \ln k + k$$

on the entropy of a random k -element subset of $[N]$, which turns out to be a very good approximation when $k \ll N$. For example, when $N = 2^{20}$ and $k = 2^8$, the approximation

$$\ln \binom{2^{20}}{2^8} \approx 2^8 \ln(2^{20}e/2^8) = 128(1 + 12 \ln 2)$$

holds with a relative error of only about 0.3%. (See ?? for a discussion of this estimate.)

3 Results for Random Dictionaries

Our first result on random dictionaries provides an

Proposition 4 *Let F and G be as above, and let $C < 2$. Then for sufficiently large N , $P(G(F(Y)) = Y) < 1/2$.*

This shows that

Proposition 5 *For $\epsilon \in (0, 1)$, let $\ln k \leq \epsilon \ln N$, and let*

$$\kappa > 2 + 4\sqrt{\epsilon} + 2\epsilon.$$

Suppose $d \geq \kappa \ln N$. Then for sufficiently large N , $G(F(Y)) = Y$ with arbitrarily high probability.

Since

$$2 + 4\sqrt{\epsilon} + 2\epsilon \leq 4(1 + \epsilon)$$

and when $k = N^\epsilon$, $4(1 + \epsilon)k \ln N = 4k \ln(N/k)$, Proposition 5 shows that.

This predictions agrees well with numerical experiments, graphed in ?. In fact, even as N varies over several orders of magnitude, the slightly weaker condition $d \geq 8k \ln N$ characterizes the regime where the set Y can be decoded with reasonably high probability by threshold decoding.

Top- k decoding performs significantly better but admits a similar “rule of thumb”: for all values of N trialed, $d = 4k \ln kN$ is very close to the smallest dimension needed for top- k decoding to succeed with high probability. See ? for an informal derivation of this bound.

There are several ways to interpret this conclusion. On one hand, it means that one-step estimates are asymptotically “inefficient” in terms of required bitrate when k is moderately large compared to N . More specifically, in a regime where N goes to infinity but $\ln k / \ln N$ converges to 1, we predict that one-step estimates require the ratio d/H to diverge to infinity.

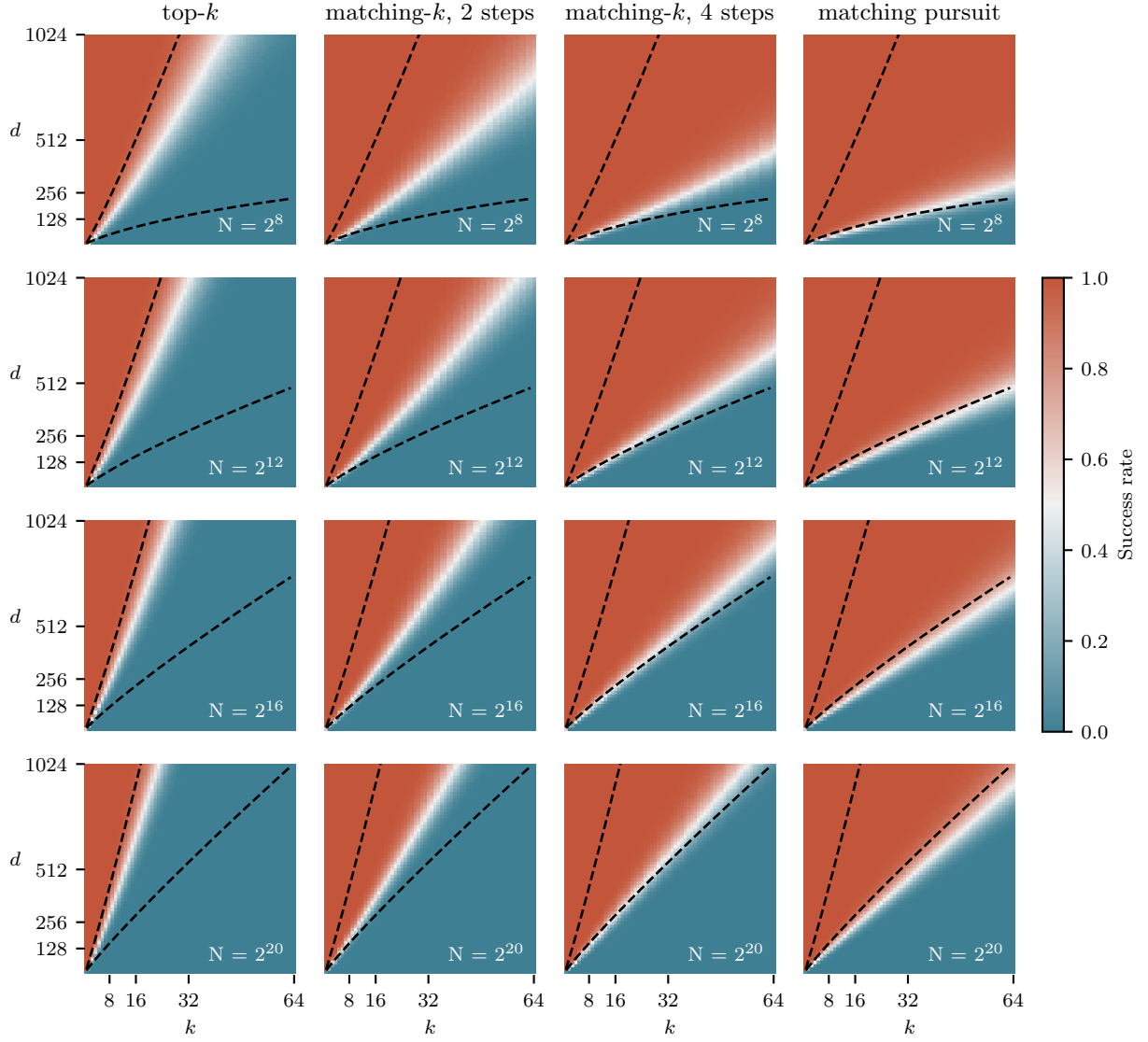


Figure 1: Empirical performance of a variety of different decoding methods at reading a k -element subset of $[N]$ from a d -dimensional superposition code with a Rademacher dictionary.

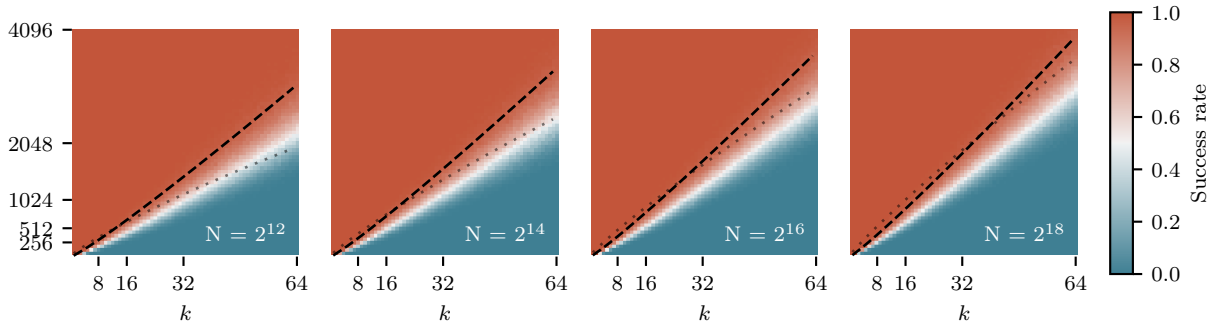


Figure 2: The prediction of Proposition 5 agrees well with numerical experiments.

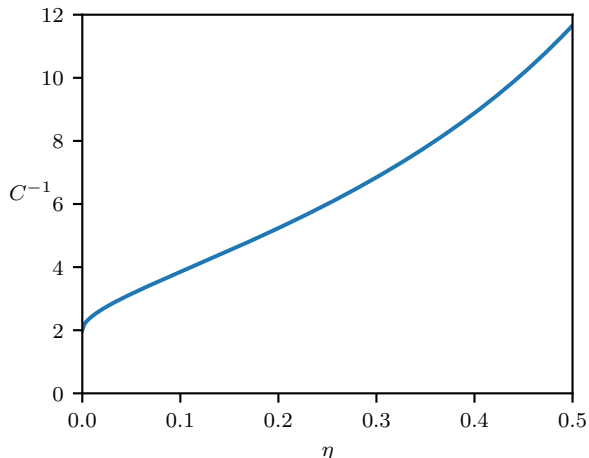


Figure 3: Inverse channel capacity (in nats per dimension) predicted by Proposition 5.

In particular, one-step estimates are asymptotically inefficient when $k/N \geq \epsilon$ for some positive ϵ . Indeed, to have $d \geq 4k \ln N$ in this case we would need $d = \Omega(N \ln(N))$, while the entropy of Y is only $O(N)$. In contrast, a hallmark result of compressive sensing implies that, when $k/N \leq \epsilon$, the vector y can be recovered from its image Fy under a random projection by a certain *convex* optimization problem so long as $d \geq \kappa(\epsilon)N$ for some constant $\kappa(\epsilon)$; for example, see Candes & Tao (2005). The failure of our one-step estimates in this particular regime is easy to prove.

On the other hand, in a sparser regime where $\ln k / \ln N < \epsilon$ for some $\epsilon < 1$, it follows from our analysis that one-step estimates are “information-efficient” in the sense that they can be decoded from superposition codes that achieve bitrates H/d larger than some positive δ . However, it is also of interest to have *non-asymptotic* information on the required bitrate. From ?? we find that one-step estimates need at least $4 \ln 2 \approx 2.8$ bits per dimension even for small k . When $k = 2^8$ and $N = 2^{20}$ this number rises to about 4.1, and the experiments of ?? show that this factor is in fact slightly optimistic. If we use threshold decoding instead of top- k , we need 8 dimensions per bit! Can other inference algorithms do significantly better?

There is an extensive literature on theory of compressive sensing. Reeves et al. (2019) shows that, in our language, superposition codes with a random dictionary are essentially optimal in the information-theoretic sense when ideal maximum-likelihood inference is used as the decoder. A series of earlier works (Joseph & Barron (2012; 2014); Rush et al. (2017)) on superposition codes also showed that, under some special conditions on y , certain decoding schemes admit bitrates up to theoretical channel capacity in the presence of Gaussian noise. However, to our knowledge, practical guarantees on the performance of iterative methods are not available for our range of k and N .

?? shows the results of a numerical experiment using an iterative method called *matching pursuit*, first suggested in Bergeaud & Mallat (1995). This is a simple “greedy” algorithm that initializes $y = 0$ and, at each of k iterations, increments the index of y whose corresponding codeword has largest inner product with $x - Fy$.

Matching pursuit far outperforms top- k decoding for the range of N and k considered earlier. When $d \geq 1.3 \log_2(eN/k)$ and $N > 2^{16}$, our experiments show that matching pursuit is very reliable. In other words, matching pursuit can reliably infer a sparse vector from just 1.3 dimensions per bit. In ?? we find that a more computationally expensive algorithm called basis pursuit can do slightly better, requiring only 0.8 dimensions per bit.

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